1) Project Name

STK Asistanı: An AI Crisis Monitor that Directs NGO Aid to Urgent and Under-Reported Crises Using Independent Data

2) Overview

STK Asistanı ("NGO Assistant") helps NGOs anywhere in the world decide **where** to run their next aid campaign, **what** it should contain and **with whom** to partner; its first pilot users are small and medium-sized NGOs in Türkiye (*STK*, hence the name), and any NGO can set its own home country and sectors. It reads independent data and news every day, classifies reports into humanitarian sectors with a self-updating machine-learning model, and ranks crises against the NGO's own sectors on an interactive world map. For each country it shows a transparent score breakdown, the trend, verified local partners (Red Crescent / Red Cross) and province-level sustainable project ideas. A **silence lens** surfaces crises that mainstream Western media ignore. An LLM (DeepSeek) is integrated to draft campaigns from verified facts only; until the API key is configured the prototype uses a template, and the live LLM is evaluated in Assignment 2.

One sentence: STK Asistanı helps NGO programme and fundraising teams find the most urgent and most overlooked crises in their sectors, using independent evidence, and turn them into campaigns, local partnerships and sustainable projects.
SDG targets: 17.17 (civil-society partnerships) · 16.1 (reduce violence-related deaths) · 2.1 (end hunger). **Success measures:** F1 against expert labels, ranking audits against domain knowledge, tests with NGO personas.

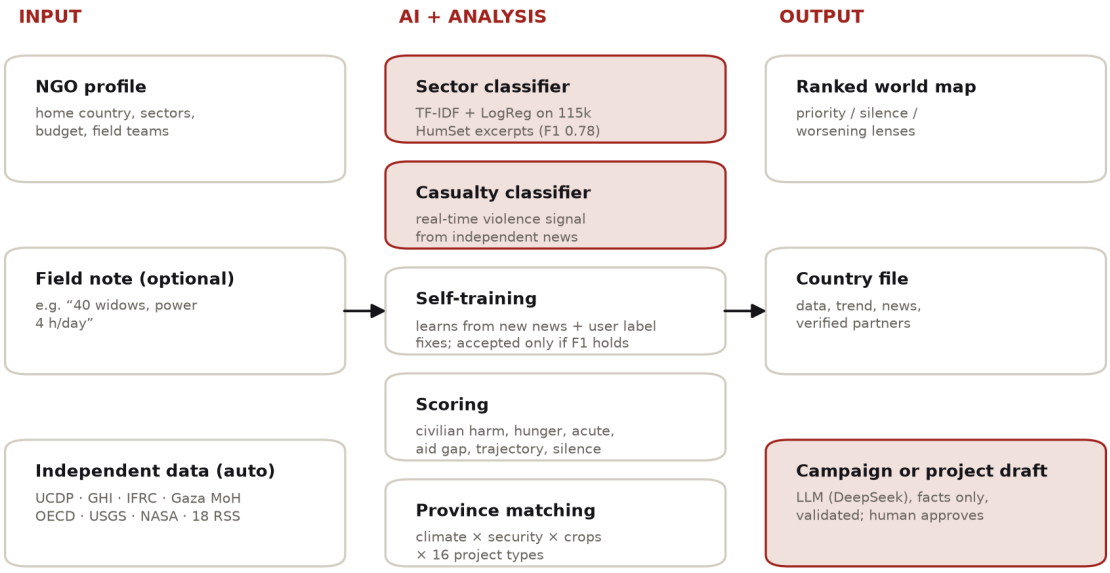


Fig. 1 — What the user puts in and what the tool produces.

3) Background

Most NGOs are small and cannot follow dozens of crises, so campaigns follow the news cycle, which is uneven. In our 30-day sample, **38% of the country-tagged world news of BBC, The Guardian and NYT concerned Ukraine and Russia, and 0% Sudan, DR Congo or Somalia**, although Sudan recorded 89,846 conflict deaths in three years (UCDP). Data infrastructure is uneven too: in a first version built on a single institutional data ecosystem, Gaza was under-ranked because of uncounted populations and conflicts not counted as acute. We therefore use multiple independent sources and never treat missing data as "no need". NLP makes automation feasible: HumSet (Fekih et al., 2022) provides 148k expert-labelled humanitarian excerpts for training.

4) Key Objectives

a) Research Questions

1. How accurately can a lightweight multilingual classifier tag live news with humanitarian sectors and casualty signals, compared with keywords, and how large is the language gap?
2. Does self-training on new news adapt the model without degrading accuracy on expert labels?

3. Can a score built only from independent sources rank acute and protracted crises credibly, including aid per person in need?
4. How large is the coverage gap between Western and independent media, and can it flag ignored crises?

b) Key Steps

1. Ingestion pipeline for 8 independent sources and 18 news feeds — *done*.
2. Sector and casualty classifiers on HumSet, self-training loop with an accuracy guard — *first version done*.
3. Country scoring, trend, aid gap, silence index; province-level project matching; web prototype — *first version done*.
4. Assignment 2: transformer and LLM zero-shot comparison; 200 hand-labelled 2026 news items; live LLM with grounding checks.
5. Assignment 3: tests with NGO personas, bias documentation, final report and video.

5) Methods and Workflow

a) Datasets

Source (publisher)	Content	Role
HumSet (DEEP / Data Friendly Space)	148,621 excerpts, EN/FR/ES, expert sector labels	classifier training
UCDP GED + 2026 candidate (Uppsala Univ.)	210,115 violent events, 2016 – Aug 2026	conflict, civilian deaths, trends
Tech for Palestine (Gaza Ministry of Health)	daily casualties; 74,392 killed, 463 starvation deaths	Palestine casualties, hunger
Global Hunger Index 2026 (Concern, WHH)	130 countries (8 provisional)	hunger, vulnerability
IFRC GO (Red Cross / Red Crescent)	164 active appeals; 220 national societies	funding gap, partners
OECD DAC2A	humanitarian aid per recipient, 2022–2024	aid per person in need
USGS, NASA EONET, Open-Meteo, Natural Earth	hazards; climate, elevation, provinces, population	acute need, project matching
18 RSS feeds (15 independent + BBC/Guardian/NYT)	news, accumulated daily	live signal, attention gap

b) Data Cleaning and Preprocessing

1. **HumSet:** 148,621 → 115,168 excerpts (no sector label or too short removed); 11 sectors mapped to 10; official split kept; class weights and per-class thresholds for a 13× imbalance.
2. **UCDP:** final and monthly releases merged; 8,478 Gaza and West Bank events re-assigned from Israel to Palestine. For Palestine the Gaza Ministry of Health is the primary source (UCDP undercounts where access is blocked).
3. **Hunger index:** "<5" → 2.5; provisional categories (e.g. Sudan, Yemen) used at the range midpoint.
4. **Aid:** Türkiye's cost of hosting Syrian refugees, reported as aid to Syria, excluded from "aid present inside Syria".
5. **News:** duplicates removed; a gazetteer tags the place where people are affected (74% of items tagged).
6. **Scaling and missing values:** deaths per 100k on a capped linear scale (skew 11; a log scale had hidden a 100-fold gap: Palestine 1,622 vs Israel 16 per 100k); missing values stay unknown or neutral, never "no need".

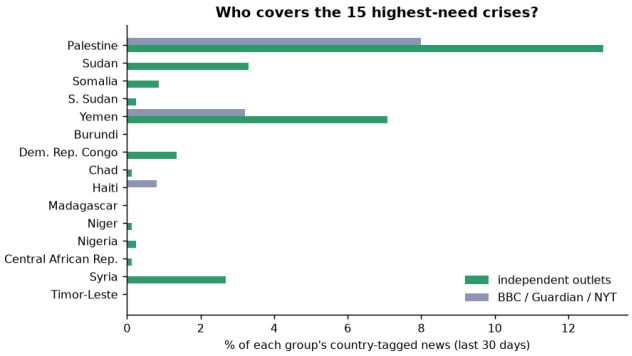


Fig. 2 — Coverage of the 15 highest-need crises.

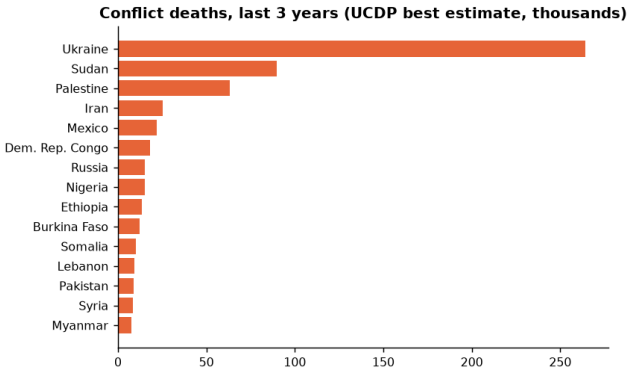


Fig. 3 — Conflict deaths, last 3 years (UCDP).

EDA insights. Ukraine, Sudan and Palestine are the deadliest conflicts; 44 countries have a "serious" or worse hunger score; Red Cross / Red Crescent appeals are 51% funded. Civilian harm and conflict intensity correlate (Spearman $\rho = 0.91$) but are both kept: they answer different questions (only 3% of recorded deaths in Ukraine are civilians). The Western news sample is still small (90 items) and grows daily.

c) Modelling

HumSet test set (12,384 excerpts)	micro-F1	macro-F1
Keyword rules (baseline)	0.558	0.539
TF-IDF + logistic regression	**0.780**	**0.740**
Casualty / displacement model	0.662	0.640

- **Choice:** TF-IDF + one-vs-rest logistic regression is fast to retrain, interpretable and multilingual. Language gap (macro-F1): EN 0.75, FR 0.72, ES 0.62.
- **Self-training:** confident news labels (≥ 0.90) and user corrections are added; a new model is accepted only if expert-test F1 holds (first run: +567 labels, 0.740 → 0.739, accepted).
- **Score:** civilian harm, conflict intensity and hunger (protracted); hazard × vulnerability, new appeals and recent violence (acute); aid per person in need; a worsening bonus (last 3 months vs previous 12). All weights are set by the user.

d) Deliverables

1. Working prototype (Python, Flask, interactive map, TR/EN) running on live data; GitHub: `github.com/tunabilgen/stk-asistani`.
2. Final dataset `data/final/country_crisis_dataset.csv` (177 × 38) with a data dictionary, and every preprocessing step with outputs in `notebooks/01_dataset_and_preprocessing.ipynb`; extended methods in `docs/concept_note_extended.pdf`.
3. Evaluation report, ethical assessment, English presentation and demo video (Assignments 2–3).

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